501_MS_AAI_Final_Project

April 12, 2026

0.1 Libraries

[11]:
```
%pip install xgboost
```

```
Requirement already satisfied: xgboost in /usr/local/lib/python3.12/dist-
packages (3.2.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from xgboost) (2.0.2)
Requirement already satisfied: nvidia-nccl-cu12 in
/usr/local/lib/python3.12/dist-packages (from xgboost) (2.29.7)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from xgboost) (1.16.3)
```

[12]:
```
import os
import kagglehub
import zipfile
import pandas as pd
import numpy as np
import scipy.stats as stats
import matplotlib.pyplot as plt
import sklearn
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import StratifiedKFold, cross_val_score
from sklearn.metrics import classification_report, confusion_matrix,␣
↪recall_score, precision_score, precision_recall_curve,␣
↪average_precision_score, f1_score
from sklearn.ensemble import GradientBoostingClassifier
from xgboost import XGBClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.ensemble import IsolationForest
```

0.2 Kaggle Authentication

[13]:
```
# Upload Kaggle API Token for Authentication
from google.colab import files
files.upload()  # upload kaggle.json
```

```
<IPython.core.display.HTML object>
```

```
Saving kaggle.json to kaggle (1).json
```

[13]:
```
{'kaggle (1).json':
b'{"username":"nixrowley","key":"KGAT_dbb3f7cb470d7d476cee2f0f9622bdf9"}\n'}
```

0.3 Exploratory Data Analysis

[14]:
```
# Download Dataset and Unzip
!kaggle datasets download -d nelgiriyewithana/
↪credit-card-fraud-detection-dataset-2023

with zipfile.ZipFile('credit-card-fraud-detection-dataset-2023.zip', 'r') as␣
↪zip_ref:
    zip_ref.extractall('/content')
```

```
Warning: Looks like you're using an outdated `kaggle` version (installed:
2.0.0), please consider upgrading to the latest version (2.0.1)
Dataset URL: https://www.kaggle.com/datasets/nelgiriyewithana/credit-card-fraud-
detection-dataset-2023
License(s): other
credit-card-fraud-detection-dataset-2023.zip: Skipping, found more recently
modified local copy (use --force to force download)
```

[15]:
```
# Load into Pandas and Inspect
df_cc_fraud = pd.read_csv('/content/creditcard_2023.csv')  # adjust filename if␣
↪needed
print(f"Original Dataset Shape: {df_cc_fraud.shape}")
df_cc_fraud.head()
```

```
Original Dataset Shape: (568630, 31)
```

[15]:
```
   id        V1        V2        V3        V4        V5        V6        V7  \
0   0 -0.260648 -0.469648  2.496266 -0.083724  0.129681  0.732898  0.519014
1   1  0.985100 -0.356045  0.558056 -0.429654  0.277140  0.428605  0.406466
2   2 -0.260272 -0.949385  1.728538 -0.457986  0.074062  1.419481  0.743511
3   3 -0.152152 -0.508959  1.746840 -1.090178  0.249486  1.143312  0.518269
4   4 -0.206820 -0.165280  1.527053 -0.448293  0.106125  0.530549  0.658849

          V8        V9  …       V21       V22       V23       V24       V25  \
0 -0.130006  0.727159  … -0.110552  0.217606 -0.134794  0.165959  0.126280
1 -0.133118  0.347452  … -0.194936 -0.605761  0.079469 -0.577395  0.190090
```

```
2 -0.095576 -0.261297  … -0.005020  0.702906  0.945045 -1.154666 -0.605564
3 -0.065130 -0.205698  … -0.146927 -0.038212 -0.214048 -1.893131  1.003963
4 -0.212660  1.049921  … -0.106984  0.729727 -0.161666  0.312561 -0.414116

        V26       V27       V28    Amount  Class
0 -0.434824 -0.081230 -0.151045  17982.10      0
1  0.296503 -0.248052 -0.064512   6531.37      0
2 -0.312895 -0.300258 -0.244718   2513.54      0
3 -0.515950 -0.165316  0.048424   5384.44      0
4  1.071126  0.023712  0.419117  14278.97      0

[5 rows x 31 columns]
```

[16]:

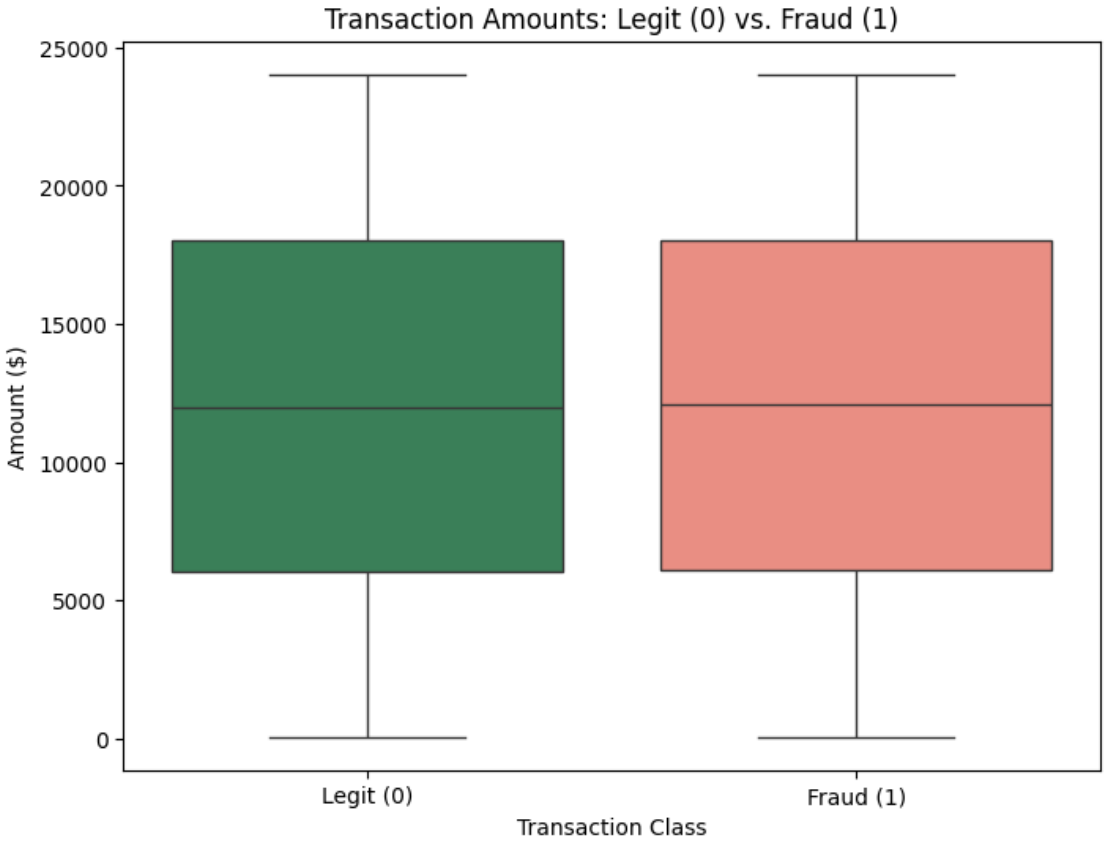
```python
# Summary of df
df_cc_fraud.info()

# Identify missing values
missing_values = df_cc_fraud.isnull().sum().max()
print(f"\nMaximum missing values in any column: {missing_values}")

# Drop non-predective columns
if 'id' in df_cc_fraud.columns:
    df_cc_fraud = df_cc_fraud.drop('id', axis=1)

# Check Class Balance
print(df_cc_fraud['Class'].value_counts(normalize=True))

# Summary Stats for Amount
amount_summary = df_cc_fraud['Amount'].describe()
print("--- Transaction Amount Range Summary ---")
print(f"Minimum: ${amount_summary['min']:.2f}")
print(f"Maximum: ${amount_summary['max']:.2f}")
print(f"Median:  ${amount_summary['50%']:.2f}")
print(f"Mean:    ${amount_summary['mean']:.2f}")
print(f"Range:   ${amount_summary['max'] - amount_summary['min']:.2f}")

amount_summary.to_excel('amount_summary_table.xlsx') # Exporting to Excel

# Define custom color mapping
custom_colors = {0: "seagreen", 1: "salmon"}

# Create the side-by-side boxplots for 'Amount' and 'Class'
plt.figure(figsize=(8, 6))
sns.boxplot(
    data=df_cc_fraud,
    x='Class',
    y='Amount',
```

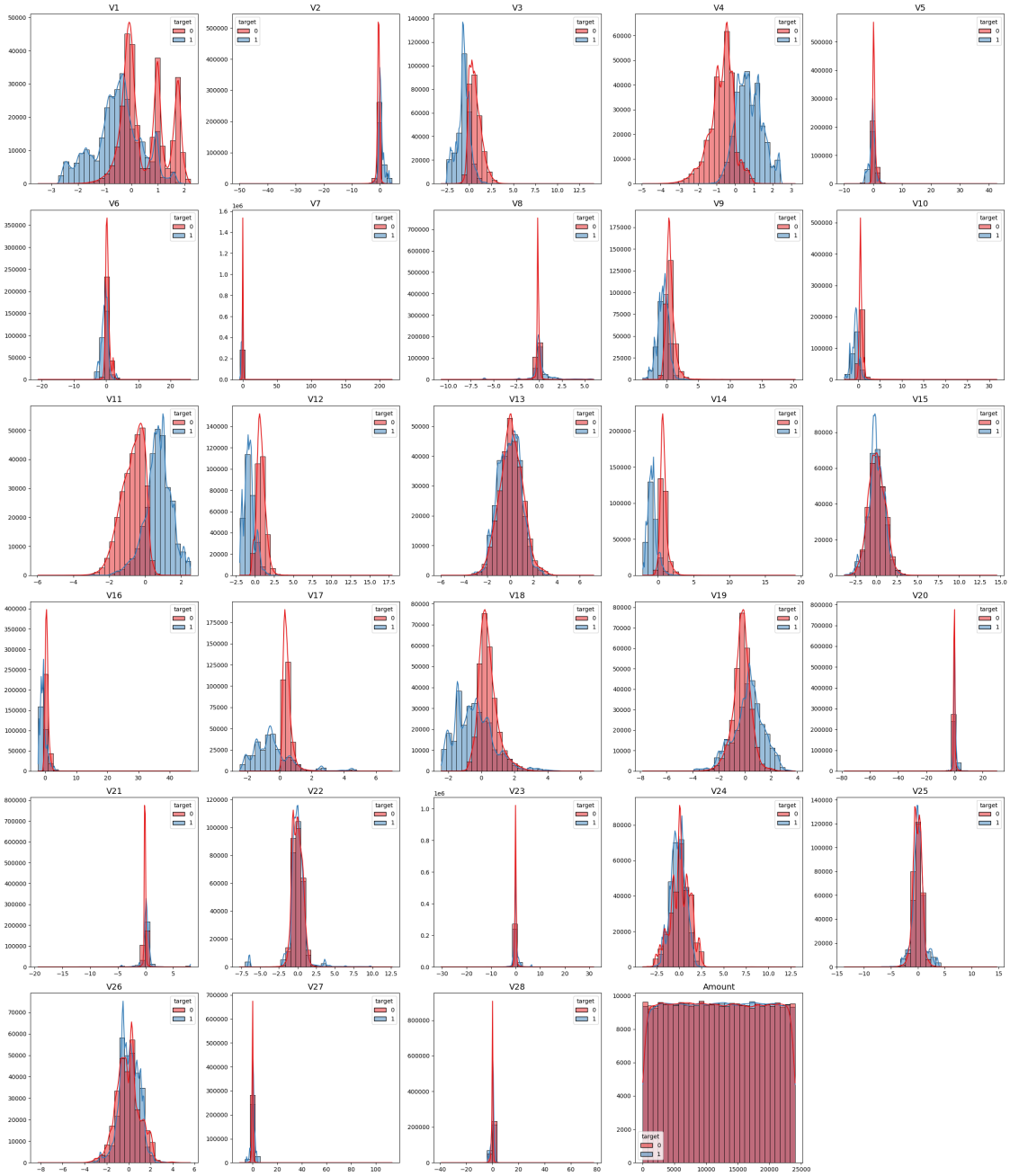
```
    hue='Class',      # This links the color to the Class
    palette=custom_colors,
    legend=False      # This prevents an unnecessary extra legend
)

# Adjust formatting for readability
plt.title('Transaction Amounts: Legit (0) vs. Fraud (1)')
plt.xlabel('Transaction Class')
plt.ylabel('Amount ($)')
plt.xticks([0, 1], ['Legit (0)', 'Fraud (1)'])

plt.savefig("boxplots.png", bbox_inches='tight', dpi=300) # Exporting to PNG

plt.show()

# Remove Target Variable and Get Features
X = df_cc_fraud.copy()
X.drop(columns=['Class'], inplace=True)
# Get all 29 continuous features
features = [col for col in X.columns]
X.head()

# Assign Target Variable
y = df_cc_fraud['Class']
y.head()

#Combine into one DataFrame for easier analysis
df = X.copy()
df['target'] = y

# Create a 6x5 grid of subplots for Histograms
fig, axes = plt.subplots(6, 5, figsize=(24, 28))
axes = axes.flatten()

# Plot features
for i, col in enumerate(features):
    sns.histplot(
        data=df,
        x=col,
        hue='target',
        kde=True,
        bins=30,
        palette='Set1',
        ax=axes[i],
        alpha=0.5,
    )
```

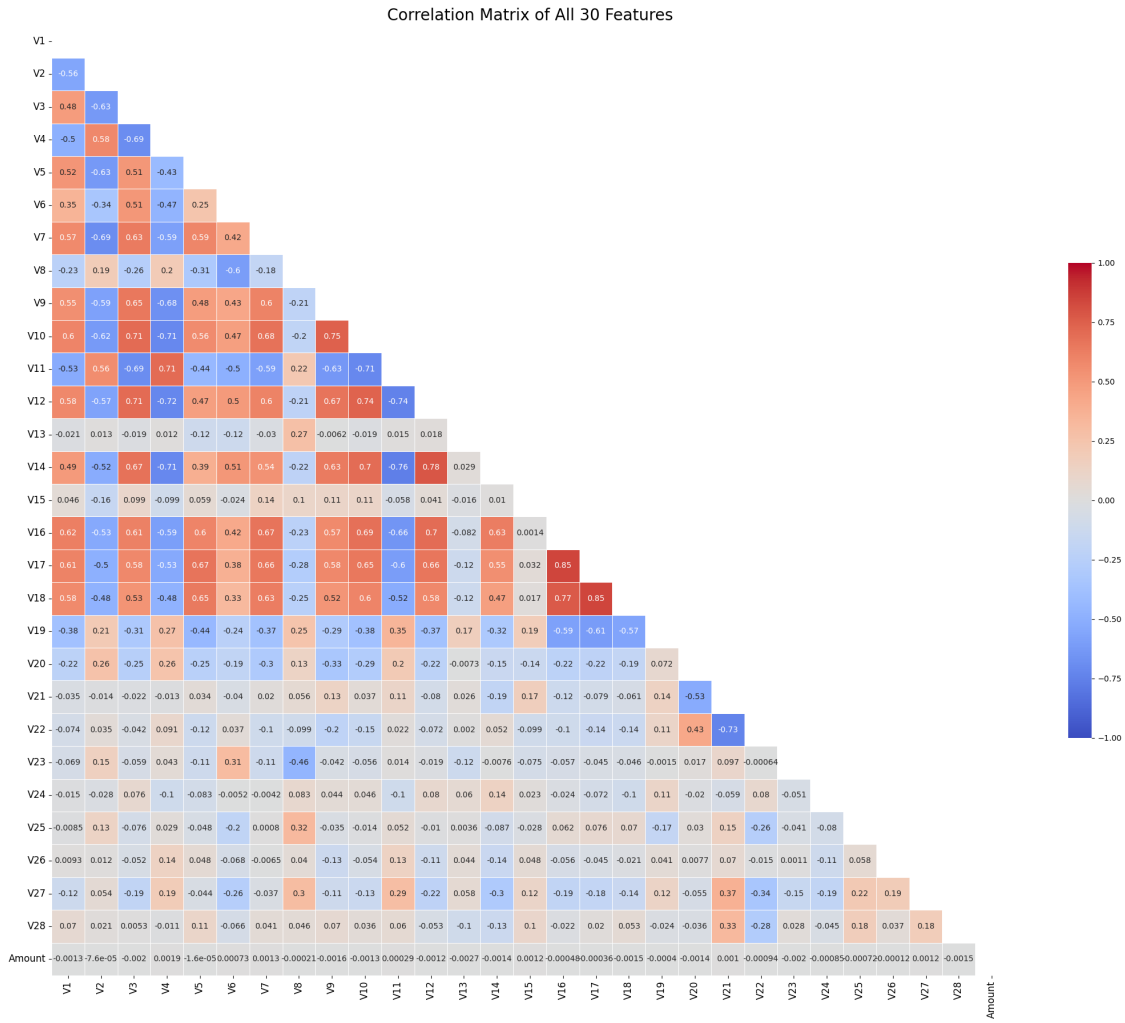
```
    axes[i].set_title(col, fontsize=14)
    axes[i].set_xlabel('')
    axes[i].set_ylabel('')

# Hide any unused subplots
for j in range(i + 1, len(axes)):
    axes[j].axis('off')

plt.tight_layout()

plt.savefig("histograms.png", bbox_inches='tight', dpi=300) # Exporting to PNG

plt.show()

# Create Heatmap of Features
corr_matrix = df[features].corr()

# Create a mask to hide the upper triangle (prevents duplicate visual␣
↪information)
mask = np.triu(np.ones_like(corr_matrix, dtype=bool))

# Set up the matplotlib figure size to accommodate 30 variables
plt.figure(figsize=(22, 18))

# Draw the heatmap
sns.heatmap(corr_matrix, mask=mask, annot=True, cmap='coolwarm', vmin=-1,␣
↪vmax=1,
            linewidths=0.5, cbar_kws={"shrink": .5})

plt.title('Correlation Matrix of All 30 Features', fontsize=20)
plt.xticks(rotation=90, fontsize=12)
plt.yticks(rotation=0, fontsize=12)
plt.tight_layout()

plt.savefig("correlation_matrix.png", bbox_inches='tight', dpi=300) # Exporting␣
↪to PNG

plt.show()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 568630 entries, 0 to 568629
Data columns (total 31 columns):
 #   Column  Non-Null Count   Dtype
---  ------  --------------   -----
 0   id      568630 non-null  int64
 1   V1      568630 non-null  float64
 2   V2      568630 non-null  float64
```

```
 3   V3      568630 non-null  float64
 4   V4      568630 non-null  float64
 5   V5      568630 non-null  float64
 6   V6      568630 non-null  float64
 7   V7      568630 non-null  float64
 8   V8      568630 non-null  float64
 9   V9      568630 non-null  float64
 10  V10     568630 non-null  float64
 11  V11     568630 non-null  float64
 12  V12     568630 non-null  float64
 13  V13     568630 non-null  float64
 14  V14     568630 non-null  float64
 15  V15     568630 non-null  float64
 16  V16     568630 non-null  float64
 17  V17     568630 non-null  float64
 18  V18     568630 non-null  float64
 19  V19     568630 non-null  float64
 20  V20     568630 non-null  float64
 21  V21     568630 non-null  float64
 22  V22     568630 non-null  float64
 23  V23     568630 non-null  float64
 24  V24     568630 non-null  float64
 25  V25     568630 non-null  float64
 26  V26     568630 non-null  float64
 27  V27     568630 non-null  float64
 28  V28     568630 non-null  float64
 29  Amount  568630 non-null  float64
 30  Class   568630 non-null  int64
dtypes: float64(29), int64(2)
memory usage: 134.5 MB

Maximum missing values in any column: 0
Class
0    0.5
1    0.5
Name: proportion, dtype: float64
--- Transaction Amount Range Summary ---
Minimum: $50.01
Maximum: $24039.93
Median:  $12030.15
Mean:    $12041.96
Range:   $23989.92
```


Transaction Amounts: Legit (0) vs. Fraud (1)
25000
20000
15000
10000
5000
0
Amount ($)
Legit (0)
Fraud (1)
Transaction Class


V1
V2
V3
V4
V5
V6
V7
V8
V9
V10
V11
V12
V13
V14
V15
V16
V17
V18
V19
V20
V21
V22
V23
V24
V25
V26
V27
V28
Amount
target
0
1



0.4 Data Cleaning and Preprocessing

[17]:
```
# Downsample to 0.2% fraud for realistic imbalance
legit_df = df_cc_fraud[df_cc_fraud['Class'] == 0]
fraud_df = df_cc_fraud[df_cc_fraud['Class'] == 1]
num_fraud = int((0.002 * len(legit_df)) / 0.998)
df_final = pd.concat([legit_df, fraud_df.sample(n=num_fraud, random_state=42)]).
 ↪sample(frac=1, random_state=42)

X = df_final.drop('Class', axis=1)
y = df_final['Class']

# Single Stratified Split (80% Train, 20% Test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
 ↪random_state=42, stratify=y)
```

```
# Baseline Scaling (For RF and GBM models only)
scaler = StandardScaler()
X_train_scaled = pd.DataFrame(scaler.fit_transform(X_train), columns=X_train.
 ↪columns)
X_test_scaled = pd.DataFrame(scaler.transform(X_test), columns=X_test.columns)

# Preparation for Hybrid Pipeline
iso = IsolationForest(contamination=0.002, random_state=42)
iso.fit(X_train_scaled) # Fit on scaled data for consistency

# Generate Anomaly Scores - decision_function returns the anomaly score (lower␣
 ↪= more abnormal)
X_train_scaled['Anomaly_Score'] = iso.decision_function(X_train_scaled)
X_test_scaled['Anomaly_Score'] = iso.decision_function(X_test_scaled)

# Define the StratifiedKFold for cross-validation
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

0.5 Random Forest Model Training and Testing

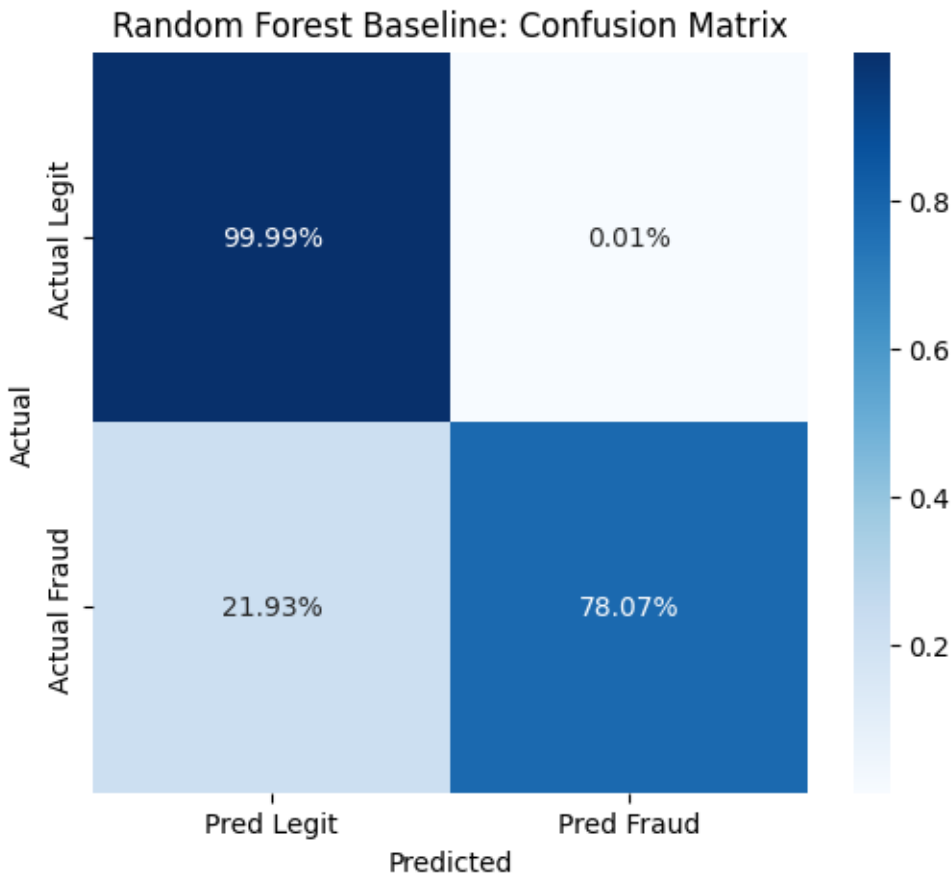
[18]:
```
# Initialize Random Forest with established parameters
# We use the scaled standard features (X_train_scaled) as the baseline␣
 ↪comparison
rf = RandomForestClassifier(n_estimators=100, max_depth=10, random_state=42,␣
 ↪n_jobs=2, verbose=1)

# Perform Stratified Cross-Validation to check model stability
print("Running 5-Fold Stratified CV on baseline features...")
rf_cv_scores = cross_val_score(rf, X_train_scaled, y_train, cv=skf,␣
 ↪scoring='average_precision')
print(f"Mean CV AUPRC: {rf_cv_scores.mean():.4f} (+/- {rf_cv_scores.std():.
 ↪4f})")

# Train on the full scaled training set
rf.fit(X_train_scaled, y_train)

# Predictions and Probabilities
y_pred_rf = rf.predict(X_test_scaled)
y_probs_rf = rf.predict_proba(X_test_scaled)[:, 1]

# Calculate Metrics
rf_auprc = average_precision_score(y_test, y_probs_rf)
report_rf = classification_report(y_test, y_pred_rf, output_dict=True)

print(f"Final Test AUPRC Score: {rf_auprc:.4f}")
```

```python
# Generate and Save Confusion Matrix
plt.figure(figsize=(6, 5))
cm_rf = confusion_matrix(y_test, y_pred_rf)
cmn_rf = cm_rf.astype('float') / cm_rf.sum(axis=1)[:, np.newaxis] # Normalized␣
↪for recall %

sns.heatmap(cmn_rf, annot=True, fmt='.2%', cmap='Blues')
plt.title('Random Forest Baseline: Confusion Matrix')
plt.xticks([0.5, 1.5], ['Pred Legit', 'Pred Fraud'])
plt.yticks([0.5, 1.5], ['Actual Legit', 'Actual Fraud'])
plt.xlabel('Predicted')
plt.ylabel('Actual')

plt.savefig("rf_confusion_matrix_baseline.png", bbox_inches='tight', dpi=300)

plt.show()

# Export Baseline Results to Excel
rf_results = pd.DataFrame([{
    'Model': 'Random Forest Baseline',
    'Total Test Samples': len(y_test),
    'Actual Fraud': y_test.sum(),
    'AUPRC Score': rf_auprc,
    'Mean CV AUPRC': rf_cv_scores.mean(),
    'Precision (Fraud)': report_rf['1']['precision'],
    'Recall (Fraud)': report_rf['1']['recall'],
    'F1-Score (Fraud)': report_rf['1']['f1-score']
}])
rf_results.to_excel("rf_baseline_results.xlsx", index=False)
```

```
Running 5-Fold Stratified CV on baseline features…
```

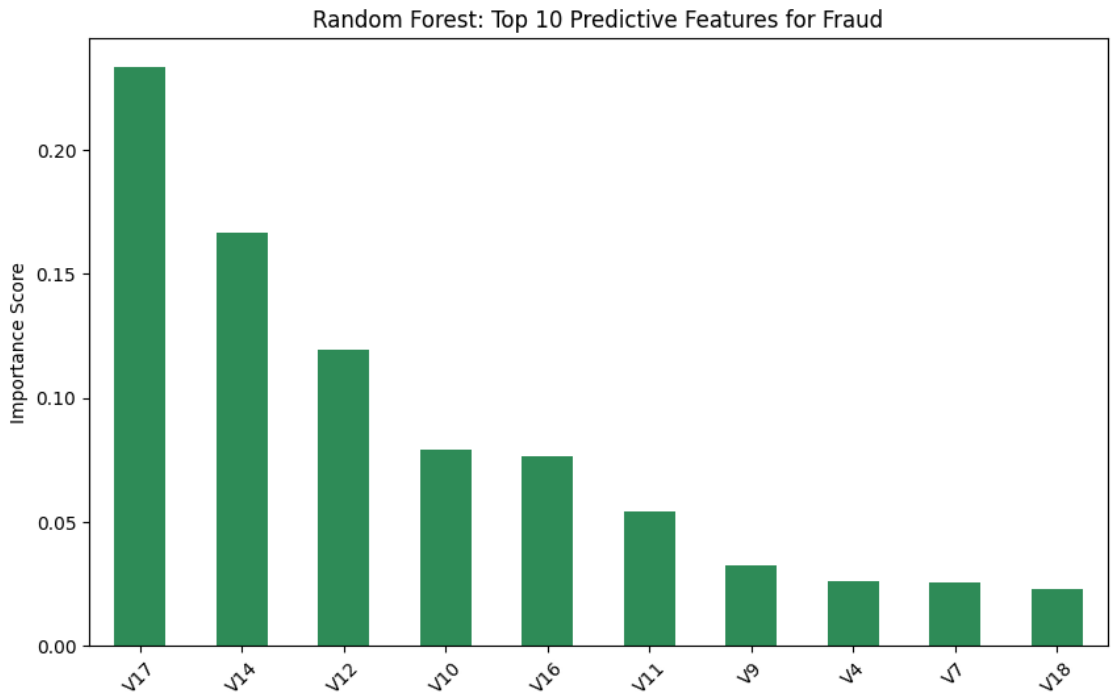
```
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:  1.1min
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:  2.3min finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:    0.1s
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:    0.2s finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:  1.1min
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:  2.3min finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:    0.2s
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:    0.4s finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:  1.0min
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:  2.2min finished
```

```
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:    0.1s
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:    0.2s finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:  1.1min
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:  2.3min finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:    0.1s
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:    0.2s finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:  1.0min
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:  2.2min finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:    0.1s
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:    0.2s finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.

Mean CV AUPRC: 0.8803 (+/- 0.0388)

[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:  1.3min
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:  2.9min finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:    0.1s
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:    0.3s finished
[Parallel(n_jobs=2)]: Using backend ThreadingBackend with 2 concurrent workers.
[Parallel(n_jobs=2)]: Done  46 tasks      | elapsed:    0.1s
[Parallel(n_jobs=2)]: Done 100 out of 100 | elapsed:    0.3s finished

Final Test AUPRC Score: 0.8727
```



0.6 Random Forest Model Feature Importance

[19]:
```
# Extract and store RF importance using the columns from the training set used
rf_importances = pd.Series(rf.feature_importances_, index=X_train_scaled.
 ↪columns)
rf_importances = rf_importances.sort_values(ascending=False)

# Plot Random Forest Feature Importance
plt.figure(figsize=(10, 6))
rf_importances.head(10).plot(kind='bar', color='seagreen')
plt.title("Random Forest: Top 10 Predictive Features for Fraud")
plt.ylabel("Importance Score")
plt.xticks(rotation=45)

plt.savefig("rf_feature_importance.png", bbox_inches='tight', dpi=300)
plt.show()
```



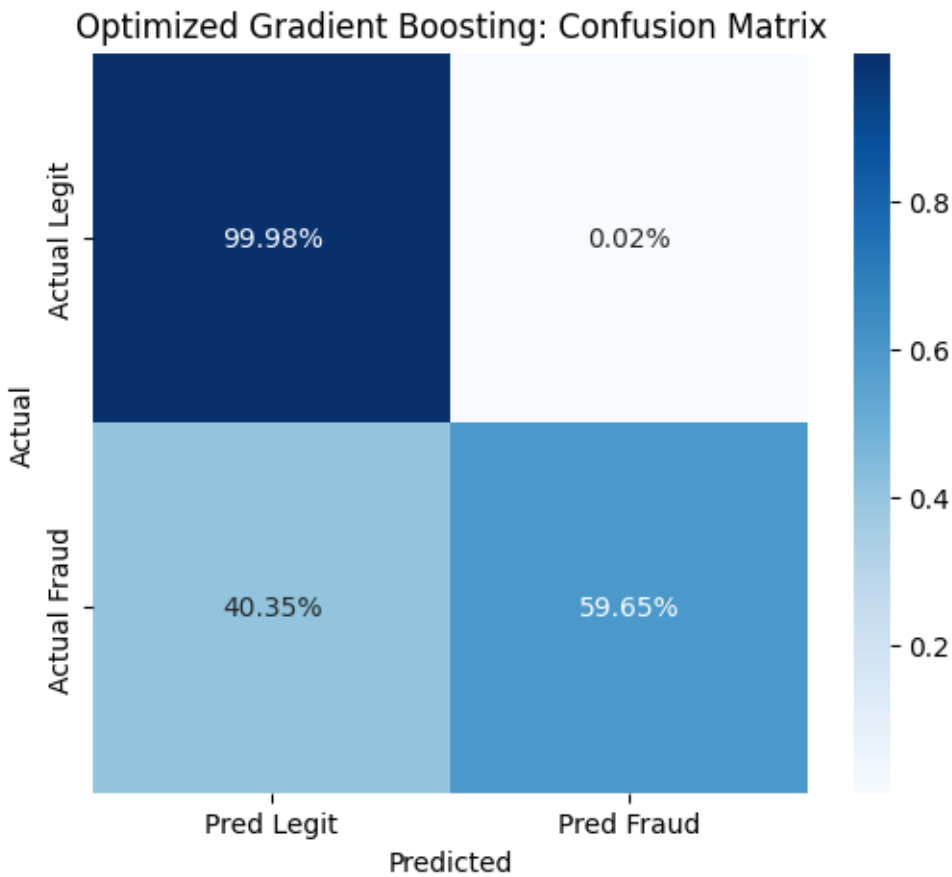
0.7 XGBoost Model Training and Optimization

[20]:
```
# Hyperparameter Optimization (Grid Search)
# Ensure 'baseline' is as strong as possible
print("Optimizing GBM hyperparameters via Grid Search...")
param_grid = {'learning_rate': [0.1], 'n_estimators': [100, 200], 'max_depth':␣
↪[3, 4]}
grid_search = GridSearchCV(
    GradientBoostingClassifier(random_state=42),
    param_grid,
    scoring='average_precision',
    cv=skf,  # Uses the StratifiedKFold defined in preprocessing
    n_jobs=-1  # Add this to use all available CPU cores
)
grid_search.fit(X_train_scaled, y_train)
gb_best = grid_search.best_estimator_
print(f"Best Parameters Found: {grid_search.best_params_}")

# Performance Validation (Cross-Validation)
print("\nRunning 5-Fold Stratified CV on optimized features...")
cv_scores = cross_val_score(gb_best, X_train_scaled, y_train, cv=skf,␣
↪scoring='average_precision')
print(f"Mean CV AUPRC: {cv_scores.mean():.4f} (+/- {cv_scores.std():.4f})")
```

```python
# Final Testing & Probability Generation
y_pred_gb = gb_best.predict(X_test_scaled)
y_probs_gb = gb_best.predict_proba(X_test_scaled)[:, 1]

# Calculate Key Metrics
gb_auprc = average_precision_score(y_test, y_probs_gb)
report_gb = classification_report(y_test, y_pred_gb, output_dict=True)
print(f"Final Test AUPRC Score: {gb_auprc:.4f}")

# Threshold Optimization
# Identifying the decision boundary that maximizes the F1-Score
precision, recall, thresholds = precision_recall_curve(y_test, y_probs_gb)
f1_scores = 2 * (precision * recall) / (precision + recall + 1e-9)
best_threshold = thresholds[np.argmax(f1_scores)]
print(f"Optimal Threshold for Optimized GB: {best_threshold:.4f}")

# Visualization: Normalized Confusion Matrix
plt.figure(figsize=(6, 5))
cm_gb = confusion_matrix(y_test, y_pred_gb)
cmn_gb = cm_gb.astype('float') / cm_gb.sum(axis=1)[:, np.newaxis]

sns.heatmap(cmn_gb, annot=True, fmt='.2%', cmap='Blues')
plt.title('Optimized Gradient Boosting: Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.xticks([0.5, 1.5], ['Pred Legit', 'Pred Fraud'])
plt.yticks([0.5, 1.5], ['Actual Legit', 'Actual Fraud'])

plt.savefig("gb_optimized_confusion_matrix.png", bbox_inches='tight', dpi=300)
plt.show()

# Export Results to Excel
# Storing metrics for the final comparison table
gb_results = pd.DataFrame([{
    'Model': 'Gradient Boosting (Optimized)',
    'Total Test Samples': len(y_test),
    'Actual Fraud': y_test.sum(),
    'AUPRC Score': gb_auprc,
    'Mean CV AUPRC': cv_scores.mean(),
    'Optimal Threshold': best_threshold,
    'Precision (Fraud)': report_gb['1']['precision'],
    'Recall (Fraud)': report_gb['1']['recall'],
    'F1-Score (Fraud)': report_gb['1']['f1-score']
}])
gb_results.to_excel("gb_optimized_results.xlsx", index=False)
```

```
Optimizing GBM hyperparameters via Grid Search…
Best Parameters Found: {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators':
```

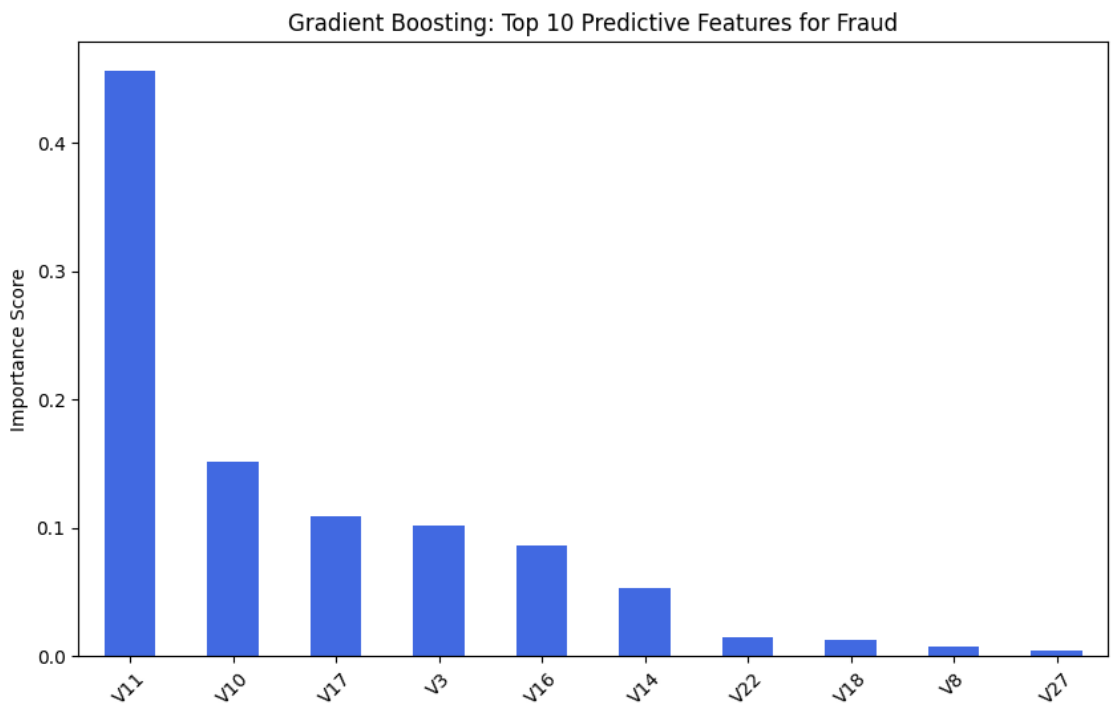
```
100}

Running 5-Fold Stratified CV on optimized features…
Mean CV AUPRC: 0.6632 (+/- 0.1264)
Final Test AUPRC Score: 0.6563
Optimal Threshold for Optimized GB: 0.2222
```



0.8 Optimized XGBoost Feature Importance

[21]:
```python
# Use the column names from the data actually passed to the model
# If X_train_scaled is a DataFrame:
if isinstance(X_train_scaled, pd.DataFrame):
    feature_names = X_train_scaled.columns
else:
    # If X_train_scaled is a NumPy array, use X_train's columns
    # BUT check if the lengths match first
    if len(gb_best.feature_importances_) == len(X_train.columns):
        feature_names = X_train.columns
    else:
```

```
        # Fallback to generic names if there is a fundamental mismatch
        feature_names = [f"Feature {i}" for i in range(len(gb_best.
↪feature_importances_))]
        print("Warning: Feature length mismatch. Check if a column was dropped/
↪added during scaling.")

gb_importances = pd.Series(gb_best.feature_importances_, index=feature_names)
gb_importances = gb_importances.sort_values(ascending=False)

# Plot Gradient Boosting Feature Importance
plt.figure(figsize=(10, 6))
gb_importances.head(10).plot(kind='bar', color='royalblue')
plt.title("Gradient Boosting: Top 10 Predictive Features for Fraud")
plt.ylabel("Importance Score")
plt.xticks(rotation=45)

plt.savefig("gb_feature_importance_optimized.png", bbox_inches='tight', dpi=300)
plt.show()
```

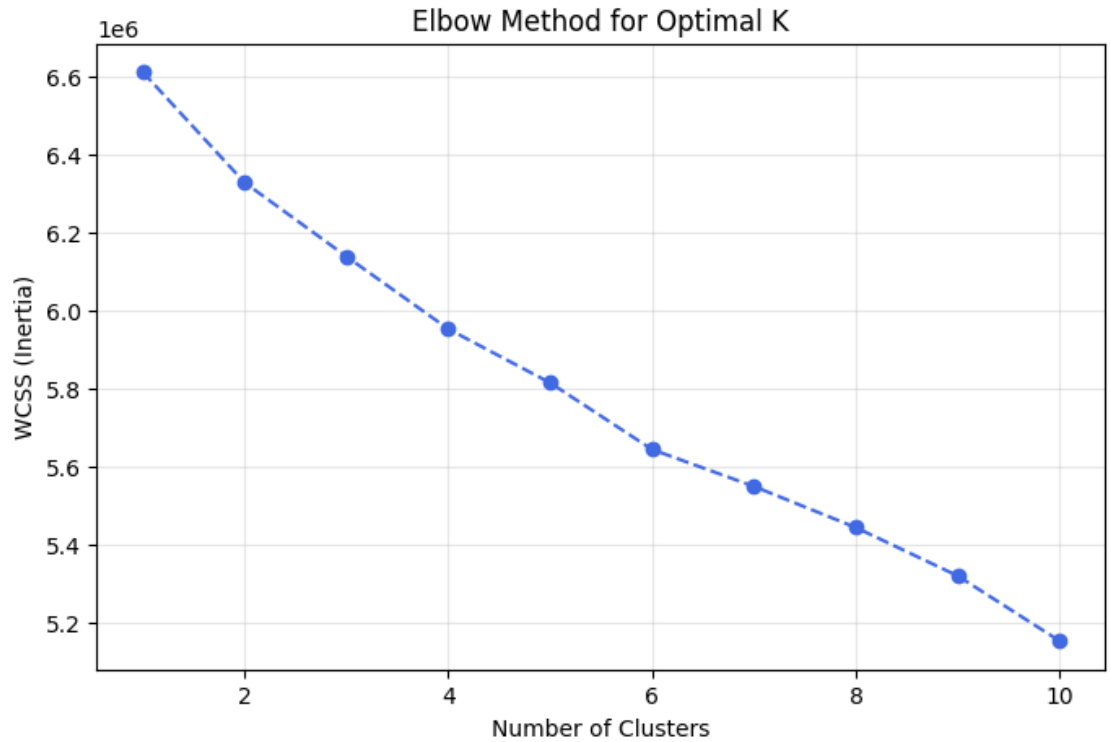


0.9 K-Means Clustering Analysis

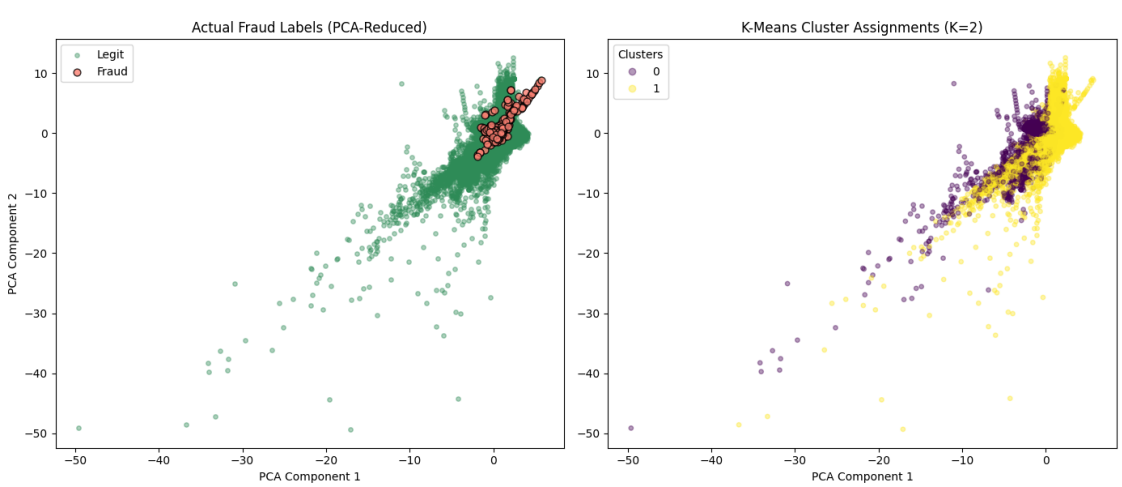
[22]:
```
# Use the training set for consistency with the model analysis flow
# K-Means is unsupervised, but we use y_train later to see if clusters align␣
↪with fraud
X_km = X_train.copy()
y_km = y_train.copy()

# Feature Scaling (K-Means is distance-based, so ALL features need scaling)
# We use the scaled training data created in the Preprocessing section
X_scaled_km = X_train_scaled.values

# Finding the Optimal K (Elbow Method)
print("Calculating Elbow Method to find optimal clusters...")
wcss = []
k_range = range(1, 11)
for i in k_range:
    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42, n_init=10)
    kmeans.fit(X_scaled_km)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(8, 5))
plt.plot(k_range, wcss, marker='o', linestyle='--', color='royalblue')
plt.title('Elbow Method for Optimal K')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS (Inertia)')
plt.grid(True, alpha=0.3)

plt.savefig('kmeans_elbow_plot.png', dpi=300, bbox_inches='tight')
plt.show()

# Fit K-Means using K=2 to see if it naturally separates Fraud vs Legit
print("Fitting K-Means with K=2...")
kmeans = KMeans(n_clusters=2, init='k-means++', random_state=42, n_init=10)
clusters = kmeans.fit_predict(X_scaled_km)

# Evaluate Cluster-to-Class Alignment
km_results_df = pd.DataFrame({'Actual_Class': y_km, 'Cluster': clusters})
print("\nCluster vs. Actual Class Distribution:")
cross_tab = pd.crosstab(km_results_df['Actual_Class'], km_results_df['Cluster'],
                        margins=True, margins_name="Total")
print(cross_tab)

# Exporting the crosstab to Excel
cross_tab.to_excel("kmeans_class_distribution.xlsx")

# Visualization using PCA (Reducing 30 features to 2D for plotting)
```

```python
print("Generating PCA for cluster visualization...")
pca_km = PCA(n_components=2)
X_pca = pca_km.fit_transform(X_scaled_km)

plt.figure(figsize=(14, 6))

# Plot 1: Actual Classes
plt.subplot(1, 2, 1)
plt.scatter(X_pca[y_km == 0, 0], X_pca[y_km == 0, 1], c='seagreen',␣
↪label='Legit', alpha=0.4, s=15)
plt.scatter(X_pca[y_km == 1, 0], X_pca[y_km == 1, 1], c='salmon',␣
↪label='Fraud', alpha=0.8, s=40, edgecolors='black')
plt.title('Actual Fraud Labels (PCA-Reduced)')
plt.legend()
plt.xlabel('PCA Component 1')
plt.ylabel('PCA Component 2')

# Plot 2: K-Means Clusters
plt.subplot(1, 2, 2)
scatter = plt.scatter(X_pca[:, 0], X_pca[:, 1], c=clusters, cmap='viridis',␣
↪alpha=0.4, s=15)
plt.title('K-Means Cluster Assignments (K=2)')
plt.xlabel('PCA Component 1')
plt.legend(*scatter.legend_elements(), title="Clusters")

plt.tight_layout()
plt.savefig('kmeans_pca_comparison.png', dpi=300, bbox_inches='tight')
plt.show()
```

```
Calculating Elbow Method to find optimal clusters…
```




```
Fitting K-Means with K=2…

Cluster vs. Actual Class Distribution:
Cluster            0       1   Total
Actual_Class
0             110397  117055  227452
1                125     330     455
Total         110522  117385  227907
Generating PCA for cluster visualization…
```



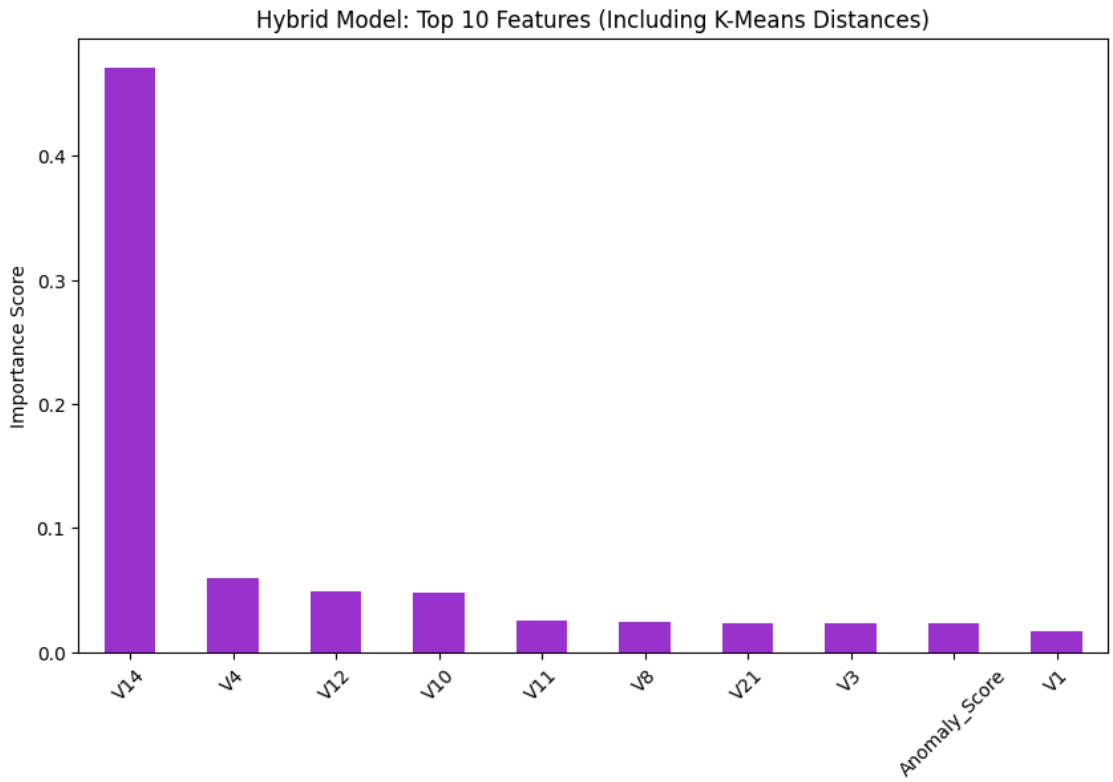
0.10 Hybrid: Gradient Boosting with K-Means Feature Engineering

[23]:
```
# Feature Engineering: Using K-Means to create distance-based features
# We calculate the distance of each point to the centroids of our 2 clusters
km_hybrid = KMeans(n_clusters=2, init='k-means++', random_state=42, n_init=10)
km_hybrid.fit(X_train_scaled)

# Create distance features for Training Set
train_distances = km_hybrid.transform(X_train_scaled)
X_train_hybrid = X_train_scaled.copy()
X_train_hybrid['Dist_Cluster_0'] = train_distances[:, 0]
X_train_hybrid['Dist_Cluster_1'] = train_distances[:, 1]
X_train_hybrid['Cluster_Label'] = km_hybrid.labels_

# Create distance features for Test Set
test_distances = km_hybrid.transform(X_test_scaled)
X_test_hybrid = X_test_scaled.copy()
X_test_hybrid['Dist_Cluster_0'] = test_distances[:, 0]
X_test_hybrid['Dist_Cluster_1'] = test_distances[:, 1]
X_test_hybrid['Cluster_Label'] = km_hybrid.predict(X_test_scaled)

# Initialize and Train XGBoost on the augmented feature set
xgb_hybrid = XGBClassifier(
    n_estimators=200,
    max_depth=4,
    learning_rate=0.1,
    scale_pos_weight=(len(y_train[y_train==0]) / len(y_train[y_train==1])), #␣
 ↪Handle imbalance
    random_state=42,
    use_label_encoder=False,
    eval_metric='logloss'
)

xgb_hybrid.fit(X_train_hybrid, y_train)

# Evaluation
y_probs_hybrid = xgb_hybrid.predict_proba(X_test_hybrid)[:, 1]
y_pred_hybrid = xgb_hybrid.predict(X_test_hybrid)
hybrid_auprc = average_precision_score(y_test, y_probs_hybrid)

print(f"Hybrid XGBoost + K-Means AUPRC: {hybrid_auprc:.4f}")

# Feature Importance: See if the Cluster features actually helped
```

```python
hybrid_importances = pd.Series(xgb_hybrid.feature_importances_,␣
↪index=X_train_hybrid.columns)
hybrid_importances = hybrid_importances.sort_values(ascending=False)

plt.figure(figsize=(10, 6))
hybrid_importances.head(10).plot(kind='bar', color='darkorchid')
plt.title("Hybrid Model: Top 10 Features (Including K-Means Distances)")
plt.ylabel("Importance Score")
plt.xticks(rotation=45)

plt.savefig("hybrid_feature_importance.png", bbox_inches='tight', dpi=300)
plt.show()

# Export Hybrid Results
hybrid_results = pd.DataFrame([{
    'Model': 'Hybrid XGBoost (K-Means Eng)',
    'AUPRC Score': hybrid_auprc,
    'Precision (Fraud)': precision_score(y_test, y_pred_hybrid),
    'Recall (Fraud)': recall_score(y_test, y_pred_hybrid),
    'F1-Score (Fraud)': f1_score(y_test, y_pred_hybrid)
}])
hybrid_results.to_excel("hybrid_model_results.xlsx", index=False)
```

```
/usr/local/lib/python3.12/dist-packages/xgboost/training.py:200: UserWarning:
[22:34:12] WARNING: /__w/xgboost/xgboost/src/learner.cc:782:
Parameters: { "use_label_encoder" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
```

```
Hybrid XGBoost + K-Means AUPRC: 0.9548
```

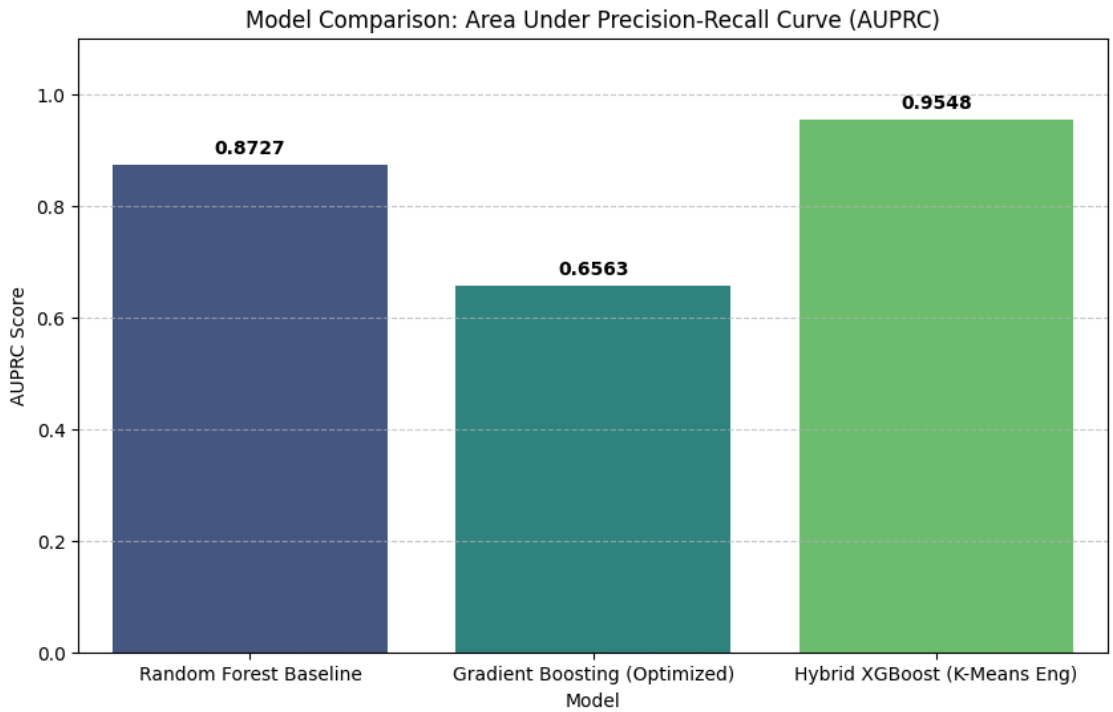


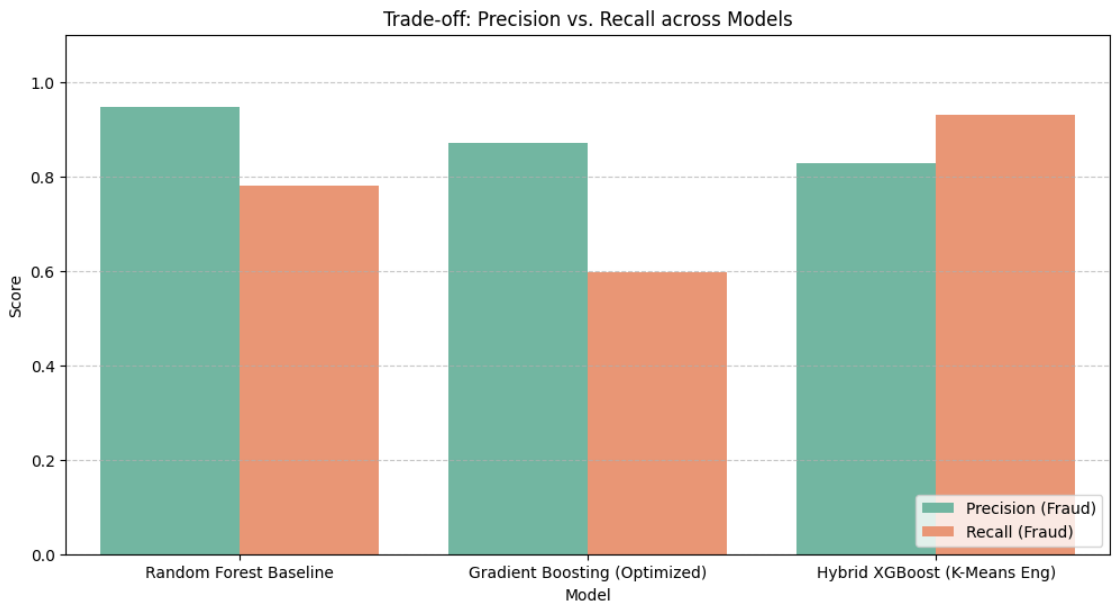
0.11 Final Performance Metrics Comparison

[24]:
```
# Consolidate results from all model DataFrames
comparison_df = pd.concat([rf_results, gb_results, hybrid_results],␣
↪ignore_index=True)

# Select the most critical columns for the executive summary
cols_to_keep = ['Model', 'AUPRC Score', 'Precision (Fraud)', 'Recall (Fraud)',␣
↪'F1-Score (Fraud)']
comparison_df = comparison_df[cols_to_keep]

# Display table
print("\nFinal Model Comparison Table:")
print(comparison_df.to_string(index=False))

# Visualization: Bar Plot Comparison of AUPRC
plt.figure(figsize=(10, 6))
sns.barplot(data=comparison_df, x='Model', y='AUPRC Score', palette='viridis',␣
↪hue='Model', legend=False)
plt.title('Model Comparison: Area Under Precision-Recall Curve (AUPRC)')
plt.ylim(0, 1.1)
```

```python
plt.ylabel('AUPRC Score')
plt.grid(axis='y', linestyle='--', alpha=0.7)

# Add data labels on top of the bars for precision
for i, score in enumerate(comparison_df['AUPRC Score']):
    plt.text(i, score + 0.02, f'{score:.4f}', ha='center', fontweight='bold')

plt.savefig("final_model_comparison_auprc.png", bbox_inches='tight', dpi=300)
plt.show()

# Visualization: Precision vs Recall Trade-off
# This shows how well each model balances "Catching Fraud" vs "Accuracy"
melted_df = comparison_df.melt(id_vars='Model', value_vars=['Precision␣
↪(Fraud)', 'Recall (Fraud)'],
                               var_name='Metric', value_name='Score')

plt.figure(figsize=(12, 6))
sns.barplot(data=melted_df, x='Model', y='Score', hue='Metric', palette='Set2')
plt.title('Trade-off: Precision vs. Recall across Models')
plt.ylim(0, 1.1)
plt.ylabel('Score')
plt.legend(loc='lower right')
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.savefig("final_precision_recall_comparison.png", bbox_inches='tight',␣
↪dpi=300)
plt.show()

# Final Export
comparison_df.to_excel("final_model_performance_report.xlsx", index=False)
print("\nFinal performance report saved to 'final_model_performance_report.
↪xlsx'")
```

```
Final Model Comparison Table:
                        Model  AUPRC Score  Precision (Fraud)  Recall (Fraud)
F1-Score (Fraud)
       Random Forest Baseline     0.872715           0.946809        0.780702
0.855769
Gradient Boosting (Optimized)     0.656258           0.871795        0.596491
0.708333
 Hybrid XGBoost (K-Means Eng)     0.954762           0.828125        0.929825
0.876033
```






```
Final performance report saved to 'final_model_performance_report.xlsx'
```